

HW 4 - Grading - rubrics (by JK, 2019-11-03)

Part 1: Basic model (29 points)

- **Section 1: making the city (6 point)**
 - 3 pts for making the function to create city
 - 1 pt for each test code
- **Section 2: making some visualization code (3 points)**
 - 2 pt for working visualization function, including selecting different colormap
 - 1 pt for testing the code
- **Section 3: Checking the Neighborhood (4 points)**
 - 1 pt for correct function onBoard
 - 3 pts for correct function getNeighborValues
- **Section 4: Starting the Rumor - The beginning of the information (2 point)**
 - 1 pt for making the city + displaying it
 - 1 pt for initializing the rumor and displaying it
- **Section 5: Spreading the rumor (14 point)**
 - 1 pt for creating and displaying the city
 - 3 pt for case with 40% density:
 - 1 pt for running the code, 1 pt for the answer
 - 1 pt for computing percentage
 - 2 pt for case with 50% density: 1 pt for running the code, 1 pt for the answer
 - 2 pt for case with 60% density: 1 pt for running the code, 1 pt for the answer
 - 4 pt for running multiple densities (5-100%): deduct points if doesn't work:
 - 1 pt for right interval
 - 2 pt for using 2 boards/cities in the loop and copy them in the begining/end of the loop
 - 1 pt for correctly computing the percentage (known/all people)
 - 2 pts for correct tipping point plot (between 50-60%)

Part 2: Improving the model (8 points)

- 1 pt answer to the question, if in direction of 'tipping point will change'
- 2 pt for correctly finished the function getNeighborValues
- 1 pt for running 1 iteration for 200x100 @ 40%
- 2 pts for running multiple densities (5-100%)
- 1 pt for answering the question about the change in tipping point
- 1 pts for correct plot for tipping point (between 37-40%)

Part 3: Wrap up (3 points)

- 1 pt for which model makes more sense in building "realistic" model
- 1 pt for scenarios where one is more appropriate than other
- 1 pt for improvements of the model; different neighborhoods, more complex rules

Part 4 (BONUS): Challenge problem (4 points)

- 1 pt for correctly finished the function getNeighborValues
 - 2 pts for finishing the code to run the simulation: including using ALPHA, BETA
 - 1 pt for printing city and percentage
-